Constraint inequalities

1

a $x \geq 0,\ y \geq 0$ b $55x + 65y \leq 800$ c $2x + 7y \leq 200$

2

a $0 \leq 5x + 8y \leq 40$ b $0 \leq 10x + 4y \leq 40$

3

a $x \geq 0,\ y \geq 0$

b $x + y \leq 200$

c The number of fluorescent light bulbs produced must be at least half as many as the number of incandescent light bulbs.

4

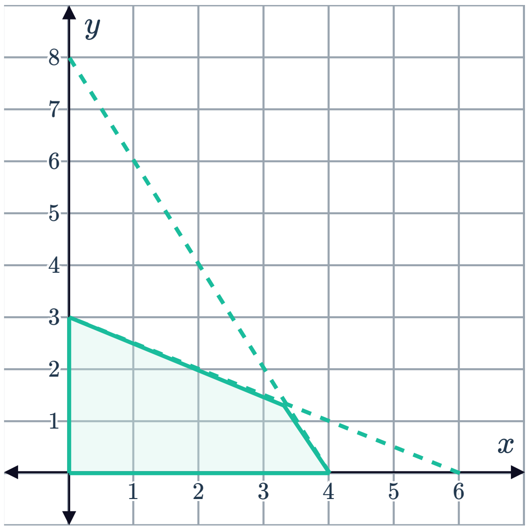
a $2x + 4y \leq 12$

b $3x + 1.5y \leq 12$

c No, x and y cannot have negative values as they are measuring number of pallets made in a day, of which the smallest value would be 0.

d $x \geq 0,\ \ y \geq 0$

e



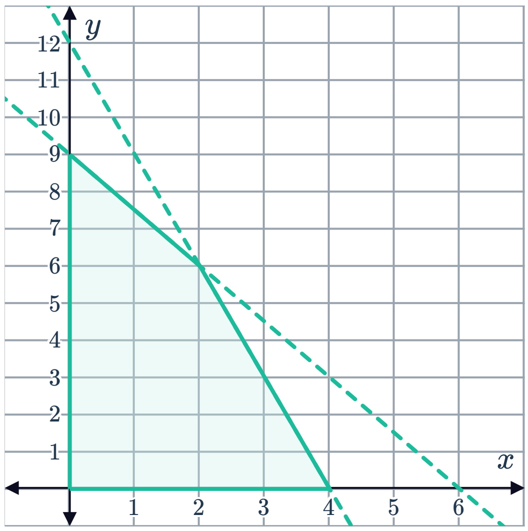
5

a $3x + 2y \leq 18$

b $4.5x + 1.5y \leq 18$

c $x \geq 0, \quad y \geq 0$

d



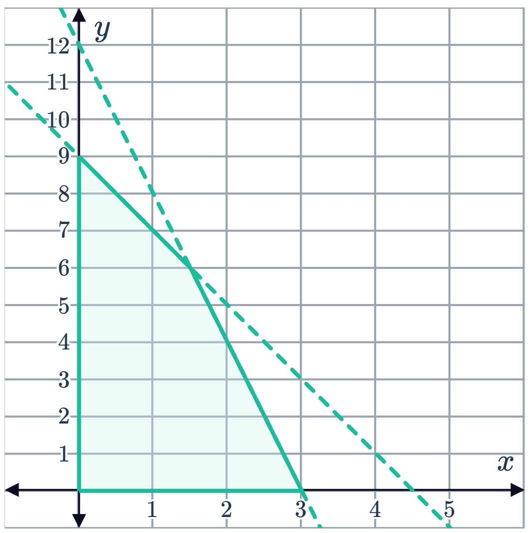
6

a $4x + 2y \leq 18$

b $6x + 1.5y \leq 18$

c $x \geq 0, \quad y \geq 0$

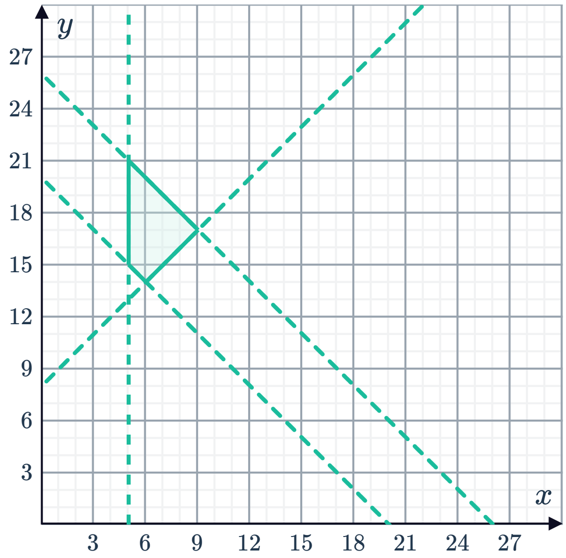
d



7

a $x \geq 5,\ y \geq x+8,\ x+y \leq 26,\ x+y \geq 20$

b



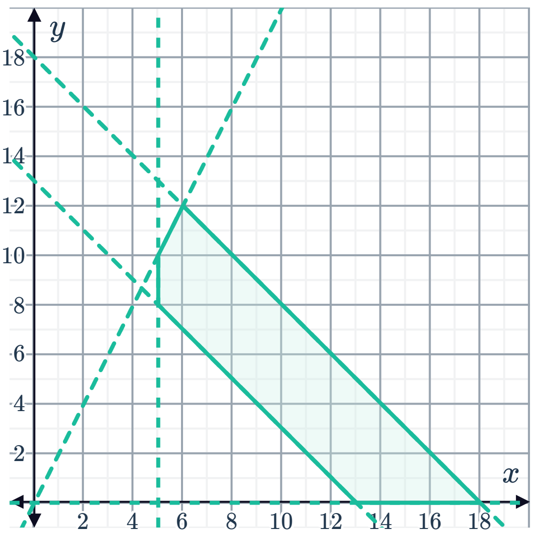
c $(5,15),(5,21),(6,14),(9,17)$

d 9 sessions

8

a $x \geq 5,\ y \geq 0,\ y \leq 2x,\ x+y \leq 18,\ x+y \geq 13$

b



c $(5,8),(5,10),(6,12),(18,0),(13,0)$

d 12 lessons

Objective functions

9

a $R = 5x + 12y$

b $R = 3x + 3.5y$

c $C = 25x + 30y$

10

a $x \geq 0,\ y \geq 0$

b $12x + 10y \leq 40$